```
#include <stdio.h>
 int main()
{ //linear search
   int i, size, target, c=0;
printf ("Enter the size Of the array: \n");
scanf ("%d", &size);

  int arr[size];
```

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```
printf ("Enter the values of indexes of array:\n");
for(i=0; i<size; i++)
{
    scanf("%d", &arr[i]);
}
printf ("Enter the value of target:\n");
scanf("%d", &target);
for (i=0; i<size; i++)
{
    if (arr[i] == target)
    {
        c = 1;
    }
}
if (c==1)
{
    printf("%d = target found", target);
}
else
{
    printf("%d = target not found!!");
}
    return 0;
}
```

O/P→ Enter the size of the array:
6
Enter the values of indexes of array:
45 6 2 3 7 789
Enter the value of target:
500
500 = target not found!!

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2. Search Product ID (Binary search)

Code:-

```
# include <stdio.h>
int main()
{
    int n, target;
    scanf("%d", &n);
    int arr[n];

    for( int i=0; i<n; i++)
    {
      scanf( "%d", &arr[i]);
    }

    scanf("%d", & target);
    int low = 0, high = n-1;
    int found = 0;

    while( low <= high)
    {
      int mid = (low + high)/2;

    if (arr[mid ] == target)
    {
      found = 1;
      break;
    }
    else if ( arr [mid] < target)
      low = mid+1;
    else
      high = mid-1;
    }
```

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```
if (found)
    printf("Product Available");
else
  printf ("Product NOT Available");

return 0;
}
```

// SORTING ALGORITHM

* ~~Bubble sorting~~

* Bubble sort :-

- Bubble sort is a sorting algorithm which is used to sort the data in ascending or descending order.
- It is not efficient for large dataset.
- The time complexity of bubble sort is $O(n^2)$
- The space complexity / Auxiliary space is $O(1)$.

code:-

```
#include <stdio.h>
void swap( int arr[], int i, int j )
{
    int temp = arr[i];
    arr [i] = arr[j];
    arr [j] = temp;
}
void bubblesort ( int arr [ ], int n )
{
    for (int i = 0; i < n-1; i ++ )
    {
        for (int j = 0; j < n-i-1 ; j++ )
        {
```

```
            if (arr[j] > arr[j+1])
            {  swap (arr, j, j+1);
            }
        }
    }
}
int main()
{
    int arr[] = { 5, 6, 1, 3 };
    int n = sizeof(arr)/sizeof(arr[0]);

    bubblesort (arr, n);

    for (int i = 0; i<n; i++)
    {  printf("%d", arr[i]);
    }
    return 0;
}
```

4. Arrange Library Book Numbers (Insertion sort)

code:-
```
#include <stdio.h>
int main()
{   int n;
    scanf("%d", &n);

    int arr[n];

    for(int i = 0; i<n; i++)
    {
        scanf("%d", & arr[i]);
    }

    for(int i=1; i<n; i++)
    {
        int key = arr[i];
        int j = i-1;

    while(j>=0 && arr[j]>key)
    {   arr[j+1] = arr[j];
        j--;
    }
```

```
        arr[j+1] = key;
    }

    for (int i= 0; i<n; i++)
    {
        printf ("%d ", arr[i]);
    }
    return 0;
}
```

5. Rank players by score (selection sort)

code:-

```
#include <stdio.h>
int main()
{
    int n;
    scanf ("%d", &n);
    int arr[n];

    for( int i= 0; i<n; i++)
    {
        scanf ("%d", &arr[i]);
    }

    for (int i=0; i<n-1; i++)
    {
        int minIndex = i;

        for (int j= i+1; j<n; j++)
        {
            if( arr[j] < arr[minIndex])
```